Physics-Based Spectral Adversarial Perturbations for Hyperspectral Vision–Language Models: A Projection-Consistent Threat Model and Evaluation Protocol

Working design note

October 15, 2026

Abstract

We develop a *physics-based* adversarial perturbation for hyperspectral imagery (HSI) that is applied to the *whole spectral cube before patchify*, and is explicitly designed to survive a learnable band-to-RGB ($B \to 3$) *adapter* used to feed a CLIP-style vision–language model (VLM) under adversarial prompt learning. The adapter (a differentiable linear/PCA/learned projection) renders a pseudo-RGB input that the frozen CLIP backbone consumes; the defense is adversarial prompt learning (CoOp-style learnable text context). Our baseline attack is a *conventional PGD-ℓ_p perturbation on the same whole cube, through the same adapter* — not an attack at the pseudo-RGB stage — so that baseline and our method differ only in *how* the perturbation is structured, not in where it is injected. Instead of an unconstrained per-band perturbation, our attack is confined to a low-dimensional, spectrally coherent manifold derived from the hyperspectral forward imaging model: smooth multiplicative illumination/transmittance changes (Beer–Lambert), additive path-radiance/scattering (Rayleigh–Mie, Ångström), and sub-pixel abundance perturbations under the linear mixing model (LMM). We prove an *adapter-survival* result: because the adapter is a low-pass $B \to 3$ projection, at matched cube budget a smooth coherent perturbation aligned with the adapter's dominant right singular vectors delivers up to a factor $\sim B/3$ more post-adapter distortion than white per-band PGD, which wastes most of its budget in the adapter's null space. This links physical plausibility directly to attack potency through the adapter. We give a fair comparison protocol and a full experiment/ablation plan against PGD-on-cube and against RGB-domain adversarial prompt-tuning defenses (AdvPT, TeCoA/ZSRobust, FAP) re-hosted in the identical pipeline, with budget matching in the (now shared) cube space, cross-attack generalization, spectral-fidelity and detectability metrics, and the ablations needed to attribute gains to the physics constraint.

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1 Setup and notation

We work with a hyperspectral reflectance cube $\boldsymbol{X} \in \mathbb{R}^{H\times W\times B}$ with B spectral bands at wavelengths $\lambda_1, \ldots, \lambda_B$. A pixel spectrum is $\boldsymbol{x}(\lambda) \in \mathbb{R}^B$. The processing pipeline that defines one training/evaluation sample is

$$\boldsymbol{X} \xrightarrow{\text{patchify (stride 1, } 13\times 13)} \{\boldsymbol{p}_i \in \mathbb{R}^{13\times 13\times B}\} \xrightarrow{\text{adapter } g_\theta\text{: } B\to 3} \text{pseudo-RGB} \xrightarrow{\text{CLIP preproc}} \text{CLIP image encoder.} \tag{1}$$

The **adapter** g_θ is a *differentiable* $B \to 3$ map (linear projection / PCA / learned linear or small MLP). It is the *only* path from the B-band cube into the 3-channel backbone input, and it is *shared by the clean forward pass and by every attack.* The defense is adversarial prompt learning (CoOp-style learnable text context) in the CLIP joint embedding space; CLIP backbones are frozen.

Both attacks live on the cube. In this revision the baseline adversary is *standard PGD on the whole cube* (an unconstrained ℓ_∞/ℓ_2 perturbation $\boldsymbol{\delta} \in \mathbb{R}^{H\times W\times B}$), passed through the *same* adapter g_θ and backbone. Our method perturbs the *same* cube but with a physics-parameterized, spectrally coherent transform. Thus baseline and ours share the attack *surface* (cube) and the *pipeline* (adapter + backbone + defense); they differ only in the *structure* of the perturbation. This is what the whole study isolates.

Why perturb the cube before patchify. With stride 1 and patch size 13, an interior pixel is shared by up to $13 \times 13 = 169$ patches. A per-patch perturbation would assign conflicting values to a shared physical pixel and is not a realizable scene perturbation. We therefore parameterize every perturbation (baseline PGD included) on the cube before patchify, so patchify (plain slicing) distributes a single consistent perturbation across all overlapping patches and autograd sums the gradient contributions back onto each cube voxel.

Definition 1 (Cube-level perturbation)**.** *The adversarial cube is* $\tilde{\boldsymbol{X}} = \mathcal{T}_{\boldsymbol{\phi}}(\boldsymbol{X})$ *with induced voxel perturbation* $\boldsymbol{\delta} = \tilde{\boldsymbol{X}} - \boldsymbol{X}$*. The baseline optimizes* $\boldsymbol{\delta}$ *freely under an* ℓ_p *ball; our method optimizes a low-dimensional* physical *parameter vector* $\boldsymbol{\phi} \in \Phi$ *and never optimizes* $\boldsymbol{\delta}$ *directly.*

2 Hyperspectral forward imaging model

The measured at-sensor spectral signal is governed by radiative transfer. For a surface of reflectance $\rho(\lambda)$ under solar illumination, a standard simplified at-sensor radiance model (cf. 6S / empirical-line form) is

$$L_s(\lambda) = \underbrace{L_p(\lambda)}_{\text{path radiance}} + \frac{T^{\uparrow}(\lambda)\, E_g(\lambda)\, \rho(\lambda)}{\pi \left(1 - \rho(\lambda)\, S(\lambda)\right)}, \tag{2}$$

where $T^{\uparrow}$ is upward transmittance, E_g the global irradiance reaching the surface, S the atmospheric spherical albedo, and L_p additive path radiance from atmospheric scattering. The global irradiance carries the downward transmittance and the solar geometry,

$$E_g(\lambda) = E_0(\lambda)\, \cos\theta_s\, T^{\downarrow}(\lambda) + E_{\text{diff}}(\lambda), \tag{3}$$

with E_0 the exo-atmospheric solar irradiance and θ_s the solar zenith angle. Transmittance obeys the Beer–Lambert law,

$$T(\lambda) = \exp\big(-\tau(\lambda)\, m\big), \qquad m \approx \frac{1}{\cos\theta_s}, \qquad \tau(\lambda) = \tau_R(\lambda) + \tau_A(\lambda) + \tau_g(\lambda), \tag{4}$$

where m is the airmass and the optical depth splits into Rayleigh (τ_R), aerosol (τ_A), and gaseous (τ_g) terms with spectral laws

$$\tau_R(\lambda) \propto \lambda^{-4}, \qquad \tau_A(\lambda) = \beta\, \lambda^{-\alpha} \quad \text{(Ångström law)}. \tag{5}$$

Path radiance L_p inherits the same λ^{-4} (molecular) and $\lambda^{-\alpha}$ (aerosol) spectral shapes.

Why this matters for the attack. Every physically meaningful perturbation of the scene — a change in sun angle, haze, aerosol load, thin cloud, adjacency, or sub-pixel material mixing — enters (2)–(5) as a *smooth, spectrally coherent* modification: multiplicative through T and E_g, additive through L_p. None is a high-frequency, band-independent noise vector of the kind PGD produces. This is the physical prior we exploit, and (Section 4) exactly the property that makes the attack survive the adapter.

3 Physics-based spectral perturbation families

Working in reflectance space (post atmospheric correction, as typical for HSI classification cubes), we model the adversarial scene as a composition of a multiplicative gain and an additive term,

$$\boxed{\tilde{\boldsymbol{x}}(\lambda) = g_{\boldsymbol{\phi}}(\lambda)\, \boldsymbol{x}(\lambda) + a_{\boldsymbol{\phi}}(\lambda)}, \tag{6}$$

applied with spatially smooth coefficients so the cube perturbation is coherent both spectrally and spatially. We instantiate $g_{\boldsymbol{\phi}}$ and $a_{\boldsymbol{\phi}}$ from the forward model.

3.1 F1 — Illumination / transmittance gain (multiplicative)

Linearizing Beer–Lambert around the nominal airmass gives a smooth multiplicative curve

$$g_{\boldsymbol{\phi}}(\lambda) = \exp\big(-\Delta\tau(\lambda)\, \Delta m\big) \cdot c, \qquad \Delta\tau(\lambda) = \Delta\beta\, \lambda^{-\alpha} + \Delta\kappa\, \lambda^{-4}, \tag{7}$$

parameters $\boldsymbol{\phi}_{F1} = \{\Delta m,\ \Delta\beta,\ \alpha,\ \Delta\kappa,\ c\}$: Δm an airmass/sun-angle shift, $\Delta\beta$ aerosol turbidity, $\Delta\kappa$ a molecular term, c a broadband illumination/shading scalar. Strictly positive and monotone-smooth in λ.

3.2 F2 — Additive scattering / path radiance

$$a_{\boldsymbol{\phi}}(\lambda) = \beta_R\,\lambda^{-4} + \beta_M\,\lambda^{-\alpha_M}, \qquad \beta_R, \beta_M \geq 0, \tag{8}$$

parameters $\boldsymbol{\phi}_{F2} = \{\beta_R, \beta_M, \alpha_M\}$. F2 raises the short wavelengths more than the near-infrared, exactly as real haze does — the physically correct "additive" counterpart to a plain ℓ_p additive budget.

3.3 F3 — Endmember / abundance perturbation (linear mixing model)

Under the LMM a pixel is a convex combination of endmember spectra, $\boldsymbol{x} = \boldsymbol{M}\boldsymbol{a} + \boldsymbol{n}$, $\boldsymbol{M} = [\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K] \in \mathbb{R}^{B\times K}$, $\boldsymbol{a} \in \Delta^{K-1}$. A physically realizable perturbation changes the sub-pixel mixing,

$$\tilde{\boldsymbol{x}} = \boldsymbol{M}(\boldsymbol{a} + \Delta\boldsymbol{a}), \qquad \mathbf{1}^\top \Delta\boldsymbol{a} = 0, \quad \boldsymbol{a} + \Delta\boldsymbol{a} \geq 0, \quad \|\Delta\boldsymbol{a}\|_2 \leq r, \tag{9}$$

so every adversarial spectrum stays inside the admissible simplex of real endmembers. $\boldsymbol{M}$ is estimated once (VCA/N-FINDR or a spectral library); the attack parameters are the abundance shifts $\boldsymbol{\phi}_{F3} = \{\Delta\boldsymbol{a}\}$.

3.4 Smoothness constraint

F1–F3 are smooth by construction. To let the optimizer use a slightly richer curve while staying physical, we optionally add a free residual $\boldsymbol{r} \in \mathbb{R}^B$ regularized by a spectral total-variation penalty,

$$\mathcal{R}_{\mathrm{sTV}}(\boldsymbol{r}) = \sum_{b=1}^{B-1} \big(r_{b+1} - r_b\big)^2 \quad \text{or} \quad \sum_{b=2}^{B-1} \big(r_{b+1} - 2r_b + r_{b-1}\big)^2. \tag{10}$$

This is the key difference from PGD: the attack is confined to the low-frequency spectral subspace real phenomena occupy — which is also the subspace the adapter preserves.

4 Adapter-survival theory: why physics beats PGD through the same adapter

Both attacks reach the backbone only through the adapter, so the quantity that matters is how much of a cube perturbation *survives* the adapter. Let $\boldsymbol{P} \in \mathbb{R}^{3\times B}$ be the (local linear) adapter with SVD $\boldsymbol{P} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$ and right singular vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3 \in \mathbb{R}^B$ spanning the *sensitive subspace* $\mathcal{S} = \mathrm{row}(\boldsymbol{P})$; its orthogonal complement $\mathcal{N} = \ker(\boldsymbol{P})$ ($\dim = B-3$) is the *blind subspace*. A cube perturbation $\boldsymbol{\delta}$ reaches the backbone as $\boldsymbol{P}\boldsymbol{\delta}$.

Lemma 1 (Optimal post-adapter perturbation)**.** *Among all spectral perturbations with $\|\boldsymbol{\delta}\|_2 \leq \epsilon$, the one maximizing the post-adapter effect is $\boldsymbol{\delta}^\star = \epsilon\,\boldsymbol{v}_1$, giving $\|\boldsymbol{P}\boldsymbol{\delta}^\star\|_2 = \epsilon\,\sigma_1$. Any component of $\boldsymbol{\delta}$ in $\mathcal{N}$ contributes nothing to $\boldsymbol{P}\boldsymbol{\delta}$.*

Proof. $\|\boldsymbol{P}\boldsymbol{\delta}\|_2 = \big\|\boldsymbol{\Sigma}\boldsymbol{V}^\top\boldsymbol{\delta}\big\|_2$ is maximized over the unit ball by aligning $\boldsymbol{V}^\top\boldsymbol{\delta}$ with the top singular direction; the maximizer is $\boldsymbol{v}_1$, value $\sigma_1\epsilon$. Writing $\boldsymbol{\delta} = \boldsymbol{\delta}_{\mathcal{S}} + \boldsymbol{\delta}_{\mathcal{N}}$, $\boldsymbol{P}\boldsymbol{\delta}_{\mathcal{N}} = \mathbf{0}$. □

Proposition 1 (Coherent vs. white perturbations through the adapter)**.** *Let $\boldsymbol{P}$ be a smooth (low-pass) adapter: its rows are smooth in λ, so $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ are smooth/low-frequency. For two perturbations of equal cube energy $\|\boldsymbol{\delta}\|_2 = \epsilon$ — a coherent $\boldsymbol{\delta}_{phys}$ from F1–F3, and a white per-band $\boldsymbol{\delta}_{pgd}$*

with energy spread uniformly over the B right-singular directions (the typical PGD outcome at the ℓ_2 ball boundary) — we have

$$\mathbb{E}\,\|\boldsymbol{P}\,\boldsymbol{\delta}_{pgd}\|_2^2 \;\approx\; \frac{\sigma_1^2+\sigma_2^2+\sigma_3^2}{B}\,\epsilon^2 \;\ll\; \|\boldsymbol{P}\,\boldsymbol{\delta}_{phys}\|_2^2 \;\approx\; \textstyle\sum_{k\le 3}\langle \boldsymbol{\delta}_{phys}, \boldsymbol{v}_k\rangle^2, \tag{11}$$

because $\boldsymbol{\delta}_{phys}$ concentrates on $\{\boldsymbol{v}_k\}_{k\le 3}$ while $\boldsymbol{\delta}_{pgd}$ wastes a fraction $\approx (B-3)/B$ of its energy in $\mathcal{N}$.

Sketch. For white $\boldsymbol{\delta}_{\text{pgd}}$ with $\mathbb{E}[\boldsymbol{\delta}\boldsymbol{\delta}^\top] = \frac{\epsilon^2}{B}\boldsymbol{I}$, $\mathbb{E}\,\|\boldsymbol{P}\boldsymbol{\delta}\|_2^2 = \frac{\epsilon^2}{B}\mathrm{tr}(\boldsymbol{P}\boldsymbol{P}^\top) = \frac{\epsilon^2}{B}\sum_{k\le 3}\sigma_k^2$. The smooth physics perturbation has large inner products with the smooth top singular vectors and little energy in the high-frequency tail that lies mostly in $\mathcal{N}$, approaching the $\boldsymbol{\delta}^\star$ bound of Lemma 1. □

Remark 1 (Interpretation and the measurable prediction)**.** *At matched cube budget, the physics attack is up to $\sim B/3$ more efficient* through the adapter *than white PGD. For $B = 200$, that ceiling is $\sim 67\times$ in retained energy; in practice PGD is not perfectly white and gradient descent finds some adapter-aligned directions, so the realized gap is smaller but still large. This yields a directly measurable experiment (E2, Section 8): the* adapter-survival ratio $\|\boldsymbol{P}\boldsymbol{\delta}\|_2^2 \,/\, \|\boldsymbol{\delta}\|_2^2$ *should be markedly higher for physics than for PGD at equal cube budget.*

Design corollary (adapter-aware physics attack). We can make the physics families provably adapter-optimal by projecting their parameter gradients onto $\mathcal{S}$, or by adding an alignment term $\langle \boldsymbol{\delta}_{\boldsymbol{\phi}}, \boldsymbol{v}_1\rangle$ under the physical constraints. A *stealth* variant instead routes part of the energy into $\mathcal{N}$ so the cube changes measurably (large SAM, defeats cube-space anomaly detectors) while the pseudo-RGB barely moves — a knob PGD does not expose.

5 Threat model and optimization

Attacker knowledge. White-box over the frozen CLIP encoder, the current learned prompts, and the adapter g_θ. If g_θ is trained jointly with the defense, the attacker uses its current weights — the honest adaptive setting (ablated in E-A5).

Budget (now a shared space). Because both attacks live on the cube, the budget is defined *identically* for both: an ℓ_∞/ℓ_2 ball $\|\boldsymbol{\delta}\| \le \epsilon$ in cube space. Our method additionally keeps a spectral-angle cap $\mathrm{SAM}(\boldsymbol{x}, \tilde{\boldsymbol{x}}) \le \gamma$ for realizability, but the *comparison* budget is the common cube ϵ. This removes the cross-space budget-matching difficulty of the earlier design.

Two inner optimizers. Both attacks maximize an *inner* adversarial loss $\mathcal{L}_{\text{in}}$ (defined with the two-loss objective in §5.1); they differ only in the variable they optimize.

Baseline: PGD on the cube.

$$\boldsymbol{\delta} \leftarrow \Pi_{\|\cdot\|\le\epsilon}\Big(\boldsymbol{\delta} + \alpha\,\mathrm{sign}\big(\nabla_{\boldsymbol{\delta}}\mathcal{L}_{\text{in}}\big)\Big), \qquad \tilde{\boldsymbol{X}} = \mathrm{clip}(\boldsymbol{X}+\boldsymbol{\delta}), \tag{12}$$

the standard ℓ_∞ PGD, applied to the whole cube and pushed through the adapter.

Ours: physics-constrained optimization (replaces direct PGD). We optimize the *physics parameters* $\boldsymbol{\phi}$ by projected gradient ascent, then project into Φ:

$$\boldsymbol{\phi} \leftarrow \Pi_\Phi\Big(\boldsymbol{\phi} + \eta\,\nabla_{\boldsymbol{\phi}}\left[\mathcal{L}_{\text{in}} - \mu\,\mathcal{R}_{\text{sTV}}\right]\Big), \qquad \tilde{\boldsymbol{X}} = \mathcal{T}_{\boldsymbol{\phi}}(\boldsymbol{X}), \tag{13}$$

where Π_Φ enforces physical ranges and the F3 simplex constraints. Because $\boldsymbol{\phi}$ is low-dimensional and $\mathcal{T}_{\boldsymbol{\phi}}$ differentiable, (13) is cheap and never produces spiky non-physical spectra.

5.1 Two-loss min–max: an adversarial inner loss and a balanced outer loss

Training is a two-level (min–max) program in which *the inner and outer steps use two different loss functions*. Let $f(\cdot;\, \boldsymbol{w}_{\text{txt}}, \theta)$ denote the pipeline (1) with learnable prompt context $\boldsymbol{w}_{\text{txt}}$ (and optionally the adapter θ) as the *defender* variables. The program is

$$\min_{\boldsymbol{w}_{\text{txt}}} \mathbb{E}_{(\boldsymbol{X},y)}\, \mathcal{L}_{\text{out}}\Big(\boldsymbol{X},\ \tilde{\boldsymbol{X}}^{\star},\ y\Big) \qquad \text{s.t.} \qquad \tilde{\boldsymbol{X}}^{\star} = \arg\max_{\tilde{\boldsymbol{X}}\in\mathcal{B}(\epsilon)}\ \mathcal{L}_{\text{in}}\big(\boldsymbol{X}, \tilde{\boldsymbol{X}}, y\big), \tag{14}$$

where $\mathcal{B}(\epsilon)$ is the common cube budget set and $\mathcal{L}_{\text{in}} \neq \mathcal{L}_{\text{out}}$. The inner loss $\mathcal{L}_{\text{in}}$ is used *only* to craft the perturbation (it is an adversarial / robustness surrogate); the outer loss $\mathcal{L}_{\text{out}}$ is a normal classification objective that balances natural and adversarial terms. This separation is deliberate: the attack generator and the defender optimize for different things.

Inner loss $\mathcal{L}_{\text{in}}$ (perturbation crafting, the max**).** The inner step is purely adversarial and label/boundary-driven. We consider three forms:

$$\text{(i) logit-CE)} \quad \mathcal{L}_{\text{in}} = \text{CE}\big(f(\tilde{\boldsymbol{X}}),\, y\big), \tag{15}$$

$$\text{(ii) embedding/semantic)} \quad \mathcal{L}_{\text{in}} = -\,\text{sim}\big(z_{\text{img}}(\tilde{\boldsymbol{X}}), t_y\big) + \max_{y'\neq y} \text{sim}\big(z_{\text{img}}(\tilde{\boldsymbol{X}}), t_{y'}\big), \tag{16}$$

$$\text{(iii) TRADES-KL)} \quad \mathcal{L}_{\text{in}} = \text{KL}\big(f(\boldsymbol{X}) \,\|\, f(\tilde{\boldsymbol{X}})\big), \tag{17}$$

with z_{img} the CLIP image embedding and t_y the learned-prompt text embedding of class y. Form (i) maximizes the classification loss of the adversarial input (Madry-style); (ii) steers the image embedding away from the true class prompt and toward the hardest wrong one, directly in the CLIP joint space the prompts live in; (iii) maximizes the divergence of the adversarial prediction from the clean one (needs no label). The perturbation is obtained by T_{atk} steps of PGD (12) on $\boldsymbol{\delta}$ (baseline) or projected ascent (13) on the physics parameters $\boldsymbol{\phi}$ (ours), applied to the chosen $\mathcal{L}_{\text{in}}$; the two methods differ only in this variable.

Which inner loss to use. We recommend the **embedding/semantic loss** (16) as the default. It is the most VLM-native adversary: it attacks exactly the joint-space geometry that prompt learning optimizes, so it stresses the defense where it is actually trained, and it couples best with the physics attack, which perturbs the same embedding through the adapter. If the min–max training becomes unstable or clean accuracy drops too far, switch the inner loss to **TRADES-KL** (17): its boundary surrogate yields less aggressive, label-free perturbations and tends to give the smoothest robustness–accuracy trade-off against a balanced outer loss. Plain **logit-CE** (15) is the simplest and a good ablation baseline, but it tends to over-concentrate on already-correct samples and is the weakest of the three. In short: embedding/semantic for the strongest VLM-aligned attack, TRADES-KL for the best trade-off and training stability, logit-CE as the baseline.

Outer loss $\mathcal{L}_{\text{out}}$ (defender, the min**).** With the adversarial cube $\tilde{\boldsymbol{X}}^{\star}$ frozen (detached from the attacker's graph), the defender minimizes a *normal classification loss that balances the natural and adversarial examples*:

$$\mathcal{L}_{\text{out}} = (1-\beta)\ \underbrace{\text{CE}\big(f(\boldsymbol{X}), y\big)}_{\text{natural}} + \beta\ \underbrace{\text{CE}\big(f(\tilde{\boldsymbol{X}}^{\star}), y\big)}_{\text{adversarial}}, \qquad \beta \in [0,1]. \tag{18}$$

This is the natural–adversarial balance: β=0 is ordinary (clean) prompt learning, β=1 is vanilla adversarial training, and intermediate β keeps clean accuracy while gaining robustness. The outer step updates only the defender variables ($\boldsymbol{w}_{\text{txt}}$, optionally the adapter θ).

Why two losses. The inner loss is a *generator* objective: its only job is to produce the hardest in-budget scene, so an unbounded, boundary-seeking surrogate (CE or KL) is appropriate. The outer loss is a *learner* objective: it must keep the model accurate on clean data too, so it is the balanced classification loss (18), not the inner surrogate. Gradients from (18) flow only into the defender variables ($\boldsymbol{w}_{\text{txt}}$, optionally θ); CLIP backbones stay frozen. Alternating the inner max of $\mathcal{L}_{\text{in}}$ and the outer min of $\mathcal{L}_{\text{out}}$ over minibatches realizes the saddle-point training. The ablations vary the inner loss (15)–(17) and the outer balance β independently.

6 Integration with the cube-level min–max loop

Both attacks slot into the same stride-1 cube-level loop; only the inner generator changes.

```
# INNER (attack):  either PGD on δ [Eq.5] OR physics on φ [Eq.6]
init perturbation
for t in 1..T_atk:
  X̃ = X+δ  (baseline)   OR   T_φ(X)  (ours)  # whole cube, before patchify
  patches = patchify(X̃)  # stride 1; perturbation shared across 169-overlap
  rgb = adapter_{B→3}(patches)  # SAME adapter for both
  logits = CLIP(rgb, prompts)
  step inner variable on attack_loss (−μ R_sTV for ours)
# OUTER (defense):  adversarial prompt learning on adv (+ clean) scene
X̃ = perturbed cube.detach()
L_def = CE(f(X̃), y) + λ CE(f(X), y)   # TRADES/AdvPT-style
prompts ← prompts − ∇ L_def
```

Keep CLIP normalization inside the differentiable graph; patchify must be plain slicing (no detach); freeze CLIP backbones; only prompts (and optionally the adapter g_θ) are trainable. The clean term preserves clean accuracy.

7 Baselines and fair comparison protocol

7.1 What is compared

There are two kinds of baseline, and they answer two different questions.

(i) Attack baseline — PGD-on-cube. The direct comparator for our *attack* is standard PGD-ℓ_∞/ℓ_2 on the whole cube (12), through the same adapter. Question: at equal cube budget, which perturbation *structure* is the stronger attack and the more realizable one?

(ii) Defense baselines — RGB adversarial prompt-tuning, re-hosted. AdvPT, TeCoA/ZSRobust, and FAP are CLIP robustness methods designed on RGB with PGD. We re-host each in the identical pipeline (1) (same cube, patchify, adapter, backbone) and train each with its own recipe against the PGD-on-cube attack. Question: does training the *defense* against our physics attack, rather than against PGD, yield prompts that are more broadly robust?

- **TeCoA / ZSRobust** — text-guided contrastive adversarial training of the image side for zero-shot robustness.
- **AdvPT** — adversarial prompt tuning; robust text prompts against PGD images, frozen CLIP.

- **FAP** — few-shot adversarial prompt learning balancing natural/adversarial features.

[Verify exact venues/years before submission.]

7.2 Fairness controls

- **Shared pipeline.** Identical cube, patchify (stride 1, 13×13), adapter g_θ, CLIP backbone, and prompt-learning defense for every method. The only variable is the attack generator (i) or the defense recipe (ii).
- **Shared budget space.** Both attacks are constrained in the *same* cube ℓ_p ball; sweep a common ϵ grid and report robustness *curves*, not single points. (We additionally report the adapter-space distortion $\Delta_{\text{rgb}} = \left\| g_\theta(\tilde{\boldsymbol{X}}) - g_\theta(\boldsymbol{X}) \right\|_p$ and SAM to expose the survival effect, but the matched budget is cube ϵ.)
- **Matched compute.** Equal attack steps T_{atk}, step size schedule, restarts, and equal defense epochs across methods.
- **Adaptive-attack check.** Evaluate each trained defense under both PGD-on-cube and the physics attack (and AutoAttack-style ensemble where applicable) to avoid overstating robustness.

8 Experiments

All result cells below are *illustrative placeholders / expected trends* to be filled from runs; they encode the hypotheses, not measurements.

Datasets. Indian Pines, Pavia University, Salinas, Houston 2018 (standard disjoint train/val/test splits); optional medical/other HSI cube for the domain variant.

Models. CLIP ViT-B/16 (and RN50 for transfer); CoOp-style learnable prompt context; adapter $g_\theta \in \{\text{PCA, learned-linear, learned-MLP}\}$, default learned-linear.

Attacks. PGD-cube (ℓ_∞, T_{atk}=10/20, with restarts) and ours (F1+F2+F3, +sTV, adapter-aware variant). Common ϵ grid in cube space.

Metrics. Overall/Average Accuracy (OA/AA), Cohen's κ; robust accuracy per ϵ; adapter-survival ratio $\text{SR} = \|\boldsymbol{P}\boldsymbol{\delta}\|_2^2 \,/\, \|\boldsymbol{\delta}\|_2^2$; SAM, spectral-TV energy, F3 simplex residual; detector AUROC; attack-success rate (ASR).

E1 — Attack potency at matched cube budget

Robust accuracy of an *undefended* prompt-learned model under each attack; lower is a stronger attack.

	Clean	Robust OA (%) vs. cube budget ϵ_∞				
Attack	OA	2/255	4/255	8/255	16/255	AUC↓
PGD-cube (baseline)	*92.0*	*78*	*61*	*40*	*22*	–
Ours (physics)	*92.0*	*66*	*45*	*24*	*11*	*lower*

Hypothesis. At equal cube ϵ, the physics attack drops accuracy further (Prop. 1): more of its budget survives the adapter.

E2 — Adapter-survival measurement (validates the theory)

Directly measure how much perturbation energy reaches the backbone.

Attack	Δ_{rgb} (post-adapter)	Survival ratio SR	Energy in $\mathcal{S}$ (%)
PGD-cube	*low*	$\approx \sum \sigma_k^2 / B$	$\sim 3/B$ *scaled*
Ours	*high*	$\gg$ *PGD*	*near 100*

Hypothesis. SR(physics) $\gg$ SR(PGD) at equal cube budget; the gap tracks the $\sim B/3$ ceiling of the Remark. Plot SR vs. B (band subsampling) to show the gap widening with more bands.

E3 — Defense (adversarial prompt learning): cross-attack grid

Train the prompt-learning defense against one attack, test against both. This is the key table for the *defense* contribution.

	Robust OA (%) when tested on		
Trained (defended) against	PGD-cube	Physics	Clean OA
No defense	*40*	*24*	*92*
PGD-cube (AdvPT/TeCoA/FAP)	*70*	*41*	*90*
Physics (ours)	*66*	*68*	*90*
Physics + PGD (mixed)	*71*	*67*	*90*

Hypothesis. Training against physics generalizes to PGD better than the reverse (off-diagonal asymmetry): the physics threat covers a broader, adapter-aligned surface. Report this per defense baseline (AdvPT/TeCoA/FAP) so the asymmetry is shown to be recipe-independent.

E4 — Realizability and stealth

Attack	SAM ↓	spectral-TV ↓	F3 simplex resid. ↓	Detector AUROC ↓
PGD-cube	*high*	*high (spiky)*	*n/a*	$\to 1.0$ *(easily caught)*
Ours	*low*	*low (smooth)*	≈ 0	$\to 0.5$ *(evades)*

Hypothesis. At *equal attack effect*, the physics attack has far lower SAM/spectral-TV and evades a spectral-smoothness / RX anomaly detector that flags PGD's per-band spikes.

E5 — Transferability

Attack crafted on model/adapter A, evaluated on B. **Hypothesis:** physics perturbations, being adapter-aligned and low-frequency, transfer across adapters, backbones (ViT↔RN), scenes, and even sensors (via spectral resampling) better than PGD, whose high-frequency content is adapter-specific.

8.1 Ablations

Ablation	Variants	Tests
A1 Physics families	F1 / F2 / F3 / F1+F2+F3, ±sTV	which mechanism carries the attack
A2 Adapter-awareness	with / without $\mathcal{S}$-alignment term	Lemma 1 directly
A3 Adapter type	PCA / learned-linear / MLP	dependence on adapter smoothness
A4 Inner loss	logit-CE / embedding / TRADES-KL	which adversary trains the best defense
A5 Adaptive attacker	adapter frozen / jointly trained	honest white-box through adapter
A6 Budget norm	ℓ_∞ / ℓ_2	robustness of the claim to norm
A7 Stealth knob	energy fraction into $\mathcal{N}$	SR vs. SAM vs. detector AUROC trade-off
A8 #PGD steps / restarts	10 / 20 / 20×5	potency gap is not a PGD-tuning artifact

9 Summary of novelty

1. A *physics-parameterized* cube-level spectral attack (illumination/ transmittance, scattering/path-radiance, LMM abundance) that is a drop-in replacement for PGD-on-cube through the *same adapter*, isolating perturbation *structure* as the only variable.
2. An *adapter-survival* theory (Lemma 1, Prop. 1) with a directly measurable prediction (E2) linking spectral coherence to attack potency through a low-pass $B \to 3$ adapter.
3. A VLM-native *embedding-space* objective for HSI adversarial prompt learning.
4. A fair, shared-budget comparison and full experiment/ablation plan against PGD-on-cube and re-hosted RGB defenses (AdvPT, TeCoA/ZSRobust, FAP), with the cross-attack generalization grid as the headline defense result.

A Practical notes

- Estimate $\boldsymbol{M}$ (endmembers) once per dataset (VCA/N-FINDR or library); cache it.
- Initialize $\boldsymbol{\phi}$ near identity ($g \approx 1$, $a \approx 0$, $\Delta\boldsymbol{a} = 0$) so the attack starts from the clean scene.
- Match PGD and physics on steps, step size, and restarts (A8) so the potency gap cannot be attributed to attack tuning.
- Compute the adapter SVD per batch (or per adapter update) to get $\{\boldsymbol{v}_k, \sigma_k\}$ for the survival ratio and the $\mathcal{S}$-alignment term.
- For memory under stride-1, attack a cube *tile* per step; consistency holds within the tile.

References

[1] C. Mao et al., "Understanding Zero-Shot Adversarial Robustness for Large-Scale Models" (TeCoA / ZSRobust). [verify venue/year]

[2] J. Zhang et al., "Adversarial Prompt Tuning for Vision–Language Models" (AdvPT). [verify venue/year]

[3] Y. Zhou et al., "Few-Shot Adversarial Prompt Learning on Vision–Language Models" (FAP). [verify venue/year]

[4] K. Zhou et al., "Learning to Prompt for Vision–Language Models" (CoOp).

[5] E. Vermote et al., “Second Simulation of the Satellite Signal in the Solar Spectrum (6S)” — radiative transfer reference.

[6] N. Keshava, J. Mustard, “Spectral Unmixing” — linear mixing model reference.